

MIMIC IV SQL

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AI in Health AI 380 H
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Query 1: Percent conversion from hospital stay to ICU stay by gender, insurance type, and race

```
WITH race_bucketed AS (  
  -- Consolidate the raw race categories into more general groups  
  SELECT  
    hadm_id,  
    subject_id,  
    insurance,  
    admittime,  
    disctime,  
    CASE  
      WHEN race LIKE '%ASIAN%' THEN 'Asian'  
      WHEN race LIKE '%BLACK%' THEN 'Black'  
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'  
      WHEN race LIKE '%WHITE%' THEN 'White'  
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska  
Native'  
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native  
Hawaiian/Pacific Islander'  
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO  
ANSWER') THEN 'Unknown/Declined'  
      ELSE 'Other' -- catches MULTIPLE RACE/ETHNICITY, OTHER, PORTUGUESE, SOUTH  
AMERICAN  
    END AS race_group  
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`  
)
```

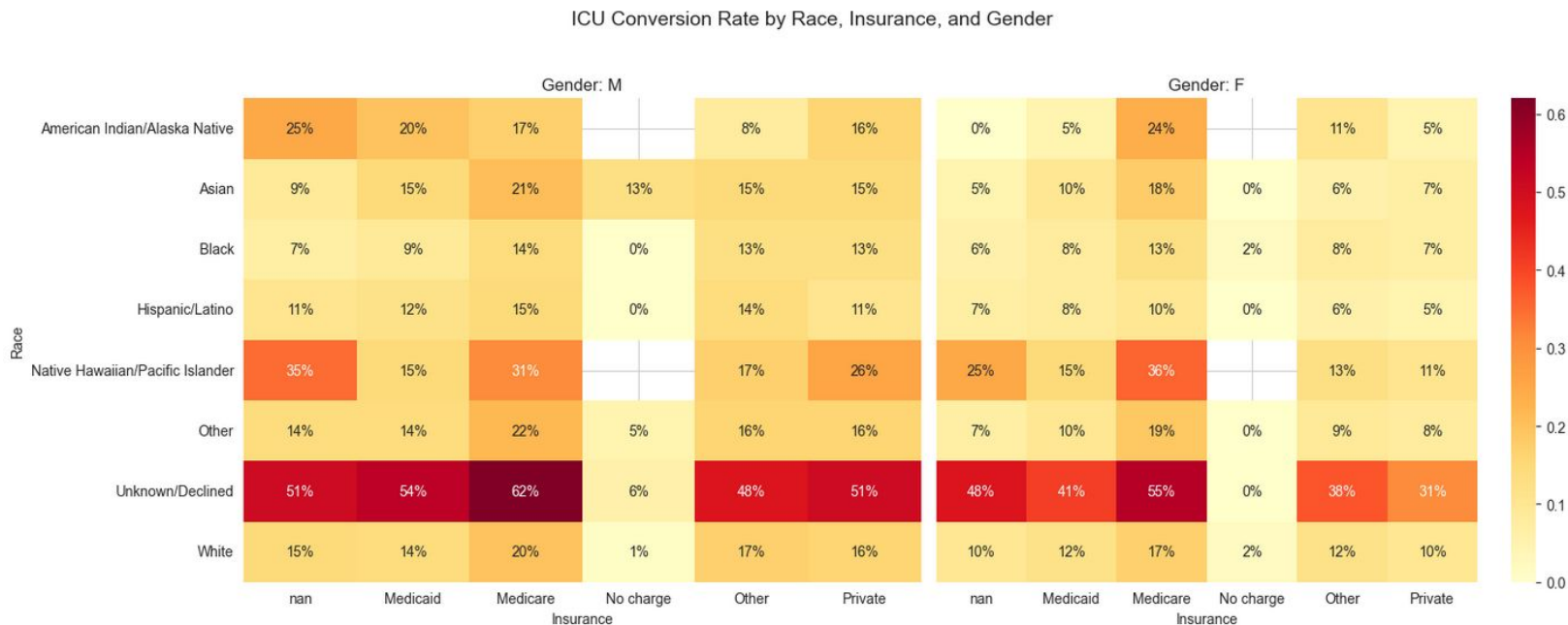
```
patient_demo AS (  
  -- Join in gender, compute length of stay in days as demographics table  
  SELECT  
    r.hadm_id,  
    r.subject_id,  
    r.insurance,  
    r.race_group,  
    p.gender  
  FROM race_bucketed r  
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p  
    ON r.subject_id = p.subject_id  
  WHERE r.disctime > r.admittime -- Filter out any unrealistic length of stays  
)
```

...

```
icu_flag AS (  
  -- Get hospital IDs from ICU table to determine if a hospital stay converted  
  SELECT DISTINCT hadm_id  
  FROM `physionet-data.mimiciv_3_1_icu.icustays`  
)  
  
admission_icu AS (  
  -- Flag each admission as ICU or not  
  SELECT  
    d.gender,  
    d.insurance,  
    d.race_group,  
    CASE WHEN i.hadm_id IS NOT NULL THEN 1 ELSE 0 END AS had_icu_stay  
  FROM patient_demo d  
  LEFT JOIN icu_flag i ON d.hadm_id = i.hadm_id  
)  
  
-- Aggregate the flags into statistics  
SELECT  
  gender,  
  insurance,  
  race_group,  
  COUNT(*) AS n_admissions,  
  SUM(had_icu_stay) AS n_icu_admissions,  
  ROUND(SUM(had_icu_stay) / COUNT(*), 2) AS icu_conversion_rate  
FROM admission_icu  
GROUP BY gender, insurance, race_group  
ORDER BY icu_conversion_rate DESC
```

Query 1: Visualization & Interpretation

Aside from unknown/declined, native hawaiians, pacific islanders, american indians, and alaskan natives had the highest ICU conversion rate by race. Medicare had the highest conversion rate by insurance type. It seems the indigenous people have a higher rate of needing critical care for some reason. I expected to see that Medicare folks would need more critical care, as they are often economically disadvantaged.



Query 1: Output

	gender	insurance	race_group	n_admissions	n_icu_admissions	icu_conversion_rate
0	M	Medicare	Unknown/Declined	4217	2619	0.62
1	F	Medicare	Unknown/Declined	3772	2059	0.55
2	M	Medicaid	Unknown/Declined	1795	976	0.54
3	M	Private	Unknown/Declined	3406	1722	0.51
4	M	None	Unknown/Declined	689	353	0.51
...	...	...	...	...	...	...
87	F	None	American Indian/Alaska Native	4	0	0.00
88	M	No charge	Black	37	0	0.00
89	F	No charge	Unknown/Declined	22	0	0.00
90	F	No charge	Asian	10	0	0.00
91	F	No charge	Other	4	0	0.00

92 rows × 6 columns

Query 2: Intensity of charting/lab events during hospital stay by gender, insurance type, and race

```
WITH race_bucketed AS (
  -- Consolidate the raw race categories into more general groups
  SELECT
    hadm_id,
    subject_id,
    insurance,
    admittime,
    disctime,
    CASE
      WHEN race LIKE '%ASIAN%' THEN 'Asian'
      WHEN race LIKE '%BLACK%' THEN 'Black'
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'
      WHEN race LIKE '%WHITE%' THEN 'White'
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska
Native'
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native
Hawaiian/Pacific Islander'
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO
ANSWER') THEN 'Unknown/Declined'
      ELSE 'Other' -- catches MULTIPLE RACE/ETHNICITY, OTHER, PORTUGUESE,
SOUTH AMERICAN
    END AS race_group
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`
),

patient_demo AS (
  -- Join in gender, compute length of stay in days as demographics table
  SELECT
    r.hadm_id,
    r.subject_id,
    r.insurance,
    r.race_group,
    p.gender,
    DATETIME_DIFF(r.disctime, r.admittime, HOUR) / 24.0 AS los_days
  FROM race_bucketed r
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p
    ON r.subject_id = p.subject_id
  WHERE r.disctime > r.admittime -- Filter out any unrealistic length of stays
),

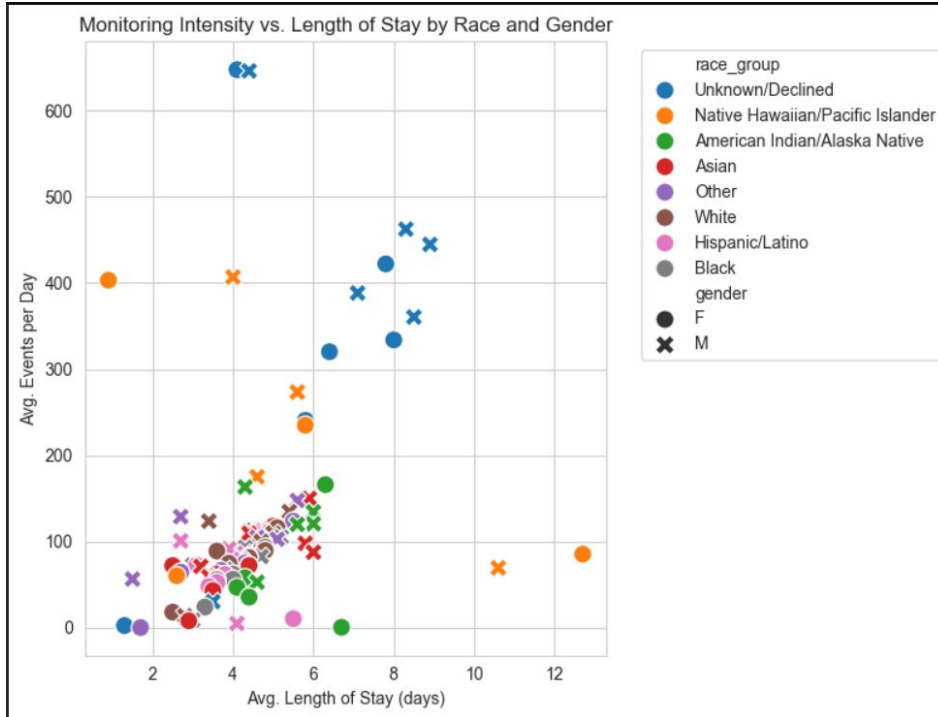
chart_counts AS (
  -- Get ICU chart events per hospital admission
  SELECT hadm_id, COUNT(*) AS chartevents
  FROM `physionet-data.mimiciv_3_1_icu.chartevents`
  GROUP BY hadm_id
),

lab_counts AS (
  -- Get lab events per hospital admission
  SELECT hadm_id, COUNT(*) AS labevents
  FROM `physionet-data.mimiciv_3_1_hosp.labevents`
  WHERE hadm_id IS NOT NULL
  GROUP BY hadm_id
),

stay_intensity AS (
  -- Combine demographics, chart count, and lab count CTEs
  SELECT
    d.hadm_id,
    d.gender,
    d.insurance,
    d.race_group,
    d.los_days,
    IFNULL(c.chartevents, 0) + IFNULL(l.labevents, 0) AS total_events,
    (IFNULL(c.chartevents, 0) + IFNULL(l.labevents, 0)) / NULLIF(d.los_days, 0) AS events_per_day
  FROM patient_demo d
  LEFT JOIN chart_counts c ON d.hadm_id = c.hadm_id
  LEFT JOIN lab_counts l ON d.hadm_id = l.hadm_id
)

-- Aggregate data into final result
SELECT
  gender,
  insurance,
  race_group,
  COUNT(*) AS n_admissions,
  ROUND(AVG(total_events), 1) AS avg_total_events,
  ROUND(AVG(events_per_day), 1) AS avg_events_per_day,
  ROUND(AVG(los_days), 1) AS avg_los_days
FROM stay_intensity
GROUP BY gender, insurance, race_group
ORDER BY avg_events_per_day DESC
```

Query 2: Visualization & Interpretation



We can see a mostly linear pattern here where events per day increases with length of stay. Thus, I can infer that monitoring intensity is relatively unbiased. I do note that native hawaiian/pacific islanders and hispanics have a lower rate of monitoring intensity compared to other races as their stay gets longer.

Query 2: Output

	gender	insurance	race_group	n_admissions	avg_total_events	avg_events_per_day	avg_los_days
0	F	None	Unknown/Declined	338	1955.8	647.7	4.1
1	M	None	Unknown/Declined	689	2698.5	646.3	4.4
2	M	Medicare	Unknown/Declined	4217	4468.4	462.5	8.3
3	M	Medicaid	Unknown/Declined	1795	4666.9	444.8	8.9
4	F	Medicare	Unknown/Declined	3772	3928.3	422.2	7.8
...	...	...	...	...	...	...	...
87	F	No charge	Asian	10	43.8	7.8	2.9
88	M	No charge	Hispanic/Latino	8	6.8	4.7	4.1
89	F	No charge	Unknown/Declined	22	7.4	2.3	1.3
90	F	None	American Indian/Alaska Native	4	5.0	0.3	6.7
91	F	No charge	Other	4	0.0	0.0	1.7

92 rows × 7 columns

Query 3: Pain medication administration by gender and race

```
WITH race_bucketed AS (
-- Consolidate the raw race categories into more general groups
SELECT
    hadm_id,
    subject_id,
    insurance,
    admittance,
    dischtime,
    CASE
        WHEN race LIKE '%ASIAN%' THEN 'Asian'
        WHEN race LIKE '%BLACK%' THEN 'Black'
        WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'
        WHEN race LIKE '%WHITE%' THEN 'White'
        WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska Native'
        WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native Hawaiian/Pacific Islander'
        WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO ANSWER') THEN 'Unknown/Declined'
        ELSE 'Other'
    END AS race_group
FROM `physionet-data.mimiciv_3_1_hosp.admissions`
),

patient_demo AS (
SELECT
    r.hadm_id,
    r.subject_id, axes = plt.subplots(1, 2, figsize=(14, 6))

metrics = [
    ('pct_received_any_pain_med', 'Any Pain Medication'),
    ('pct_received_opioid', 'Opioid Specifically')
]

for ax, (metric, title) in zip(axes, metrics):
    pivot = query3_df.pivot(index='race_group', columns='gender', values=metric)

    sns.heatmap(
        pivot,
        annot=True,
        fmt='0%',
        cmap='YlOrRd',
        ax=ax,
        cbar=True,
        vmin=0, vmax=1
    )
    ax.set_title(title)
    ax.set_xlabel("Gender")
    ax.set_ylabel("Race" if ax == axes[0] else "")

plt.suptitle("Pain Medication Administration Rate by Gender and Race", y=1.03, fontsize=14)
plt.tight_layout()
plt.show()

r.race_group,
p.gender
FROM race_bucketed r
JOIN `physionet-data.mimiciv_3_1_hosp.patients` p
ON r.subject_id = p.subject_id
WHERE r.dischtime > r.admittime
),

pain_med_orders AS (
-- Flag prescription rows matching common pain medications (opioid + non-opioid analgesics)
SELECT
    hadm_id,
    drug,
    CASE
        WHEN REGEXP_CONTAINS(UPPER(drug),
            r'MORPHINE|FENTANYL|OXYCODONE|HYDROMORPHONE|HYDROCODONE|TRAMADOL|CODEINE')
            THEN 'Opioid'
        WHEN REGEXP_CONTAINS(UPPER(drug), r'ACETAMINOPHEN|IBUPROFEN|NAPROXEN|KETOROLAC')
            THEN 'Non-opioid analgesic'
        ELSE NULL
    END AS pain_med_class
FROM `physionet-data.mimiciv_3_1_hosp.prescriptions`
),

pain_med_flagged AS (
-- Keep only rows that actually matched a pain medication
SELECT hadm_id, pain_med_class
FROM pain_med_orders
WHERE pain_med_class IS NOT NULL
),

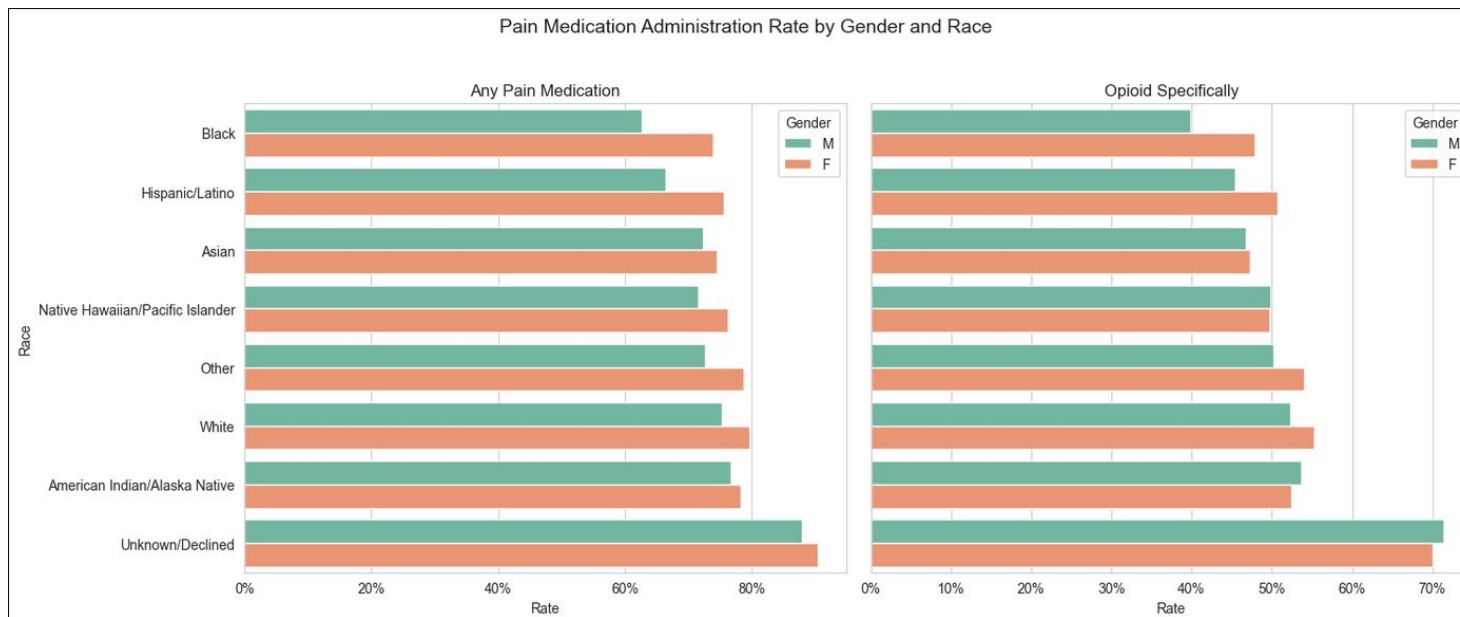
admission_pain_counts AS (
-- Count total pain med orders and opioid-specific orders per admission
SELECT
    hadm_id,
    COUNT(*) AS n_pain_med_orders,
    COUNTIF(pain_med_class = 'Opioid') AS n_opioid_orders
FROM pain_med_flagged
GROUP BY hadm_id
),

admission_pain AS (
-- Join demographics to pain med order counts
SELECT
    d.gender,
    d.race_group,
    IFNULL(c.n_pain_med_orders, 0) AS n_pain_med_orders,
    IFNULL(c.n_opioid_orders, 0) AS n_opioid_orders,
    CASE WHEN c.hadm_id IS NOT NULL THEN 1 ELSE 0 END AS received_pain_med,
    CASE WHEN c.n_opioid_orders > 0 THEN 1 ELSE 0 END AS received_opioid
FROM patient_demo d
LEFT JOIN admission_pain_counts c ON d.hadm_id = c.hadm_id
)

-- Aggregate into final result: pain med administration rate by gender and race
SELECT
    gender,
    race_group,
    COUNT(*) AS n_admissions,
    ROUND(AVG(received_pain_med), 3) AS pct_received_any_pain_med,
    ROUND(AVG(received_opioid), 3) AS pct_received_opioid,
    ROUND(AVG(n_pain_med_orders), 1) AS avg_pain_med_orders_per_admission,
    ROUND(AVG(n_opioid_orders), 1) AS avg_opioid_orders_per_admission
FROM admission_pain
GROUP BY gender, race_group
ORDER BY pct_received_opioid DESC
```

Query 3: Visualization & Interpretation

Results are pleasantly even. We can see that women are almost always prescribed more pain medication than men. Further research is needed to investigate if this caused by occurrence of childbirth. Black and hispanic women tend to get more medication than their male counterparts. It's worth doing research on whether this is by patient denial (toxic masculinity), or lack of prescription.



Query 3: Output

	gender	race_group	n_admissions	pct_received_any_pain_med	pct_received_opioid	avg_pain_med_orders_per_admission	avg_opioid_orders_per_admission
0	M	Unknown/Declined	10649	0.879	0.714	5.8	3.4
1	F	Unknown/Declined	8831	0.903	0.700	5.4	3.0
2	F	White	180105	0.796	0.552	3.8	2.1
3	F	Other	12182	0.787	0.540	3.7	1.9
4	M	American Indian/Alaska Native	567	0.767	0.536	3.7	2.2
5	F	American Indian/Alaska Native	680	0.782	0.524	3.7	1.9
6	M	White	180306	0.752	0.523	3.6	2.2
7	F	Hispanic/Latino	17428	0.755	0.507	3.3	1.7
8	M	Other	11052	0.726	0.502	3.6	2.1
9	M	Native Hawaiian/Pacific Islander	305	0.715	0.498	4.2	2.5
10	F	Native Hawaiian/Pacific Islander	189	0.762	0.497	3.4	1.7
11	F	Black	53455	0.739	0.478	3.1	1.6
12	F	Asian	11140	0.744	0.472	3.0	1.4
13	M	Asian	8603	0.723	0.467	3.2	1.8
14	M	Hispanic/Latino	14774	0.663	0.454	3.0	1.8
15	M	Black	35582	0.626	0.399	2.7	1.5

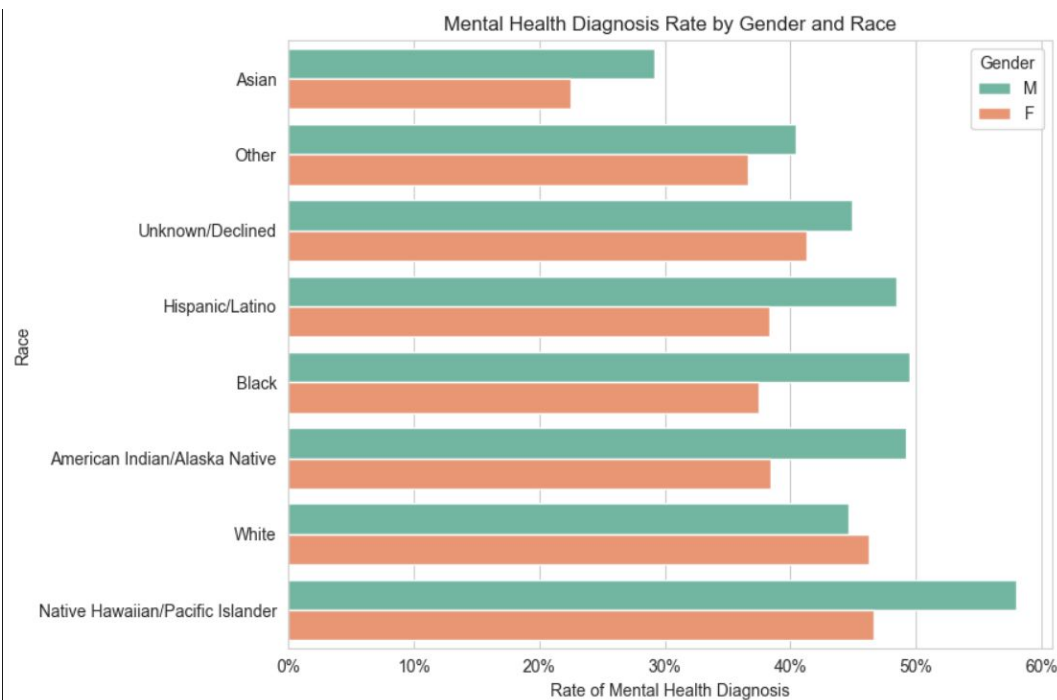
Query 4: Rate of mental health diagnosis by gender and race

```
WITH race_bucketed AS (  
  -- Consolidate the raw race categories into more general groups  
  SELECT  
    hadm_id,  
    subject_id,  
    admittime,  
    dischtime,  
    CASE  
      WHEN race LIKE '%ASIAN%' THEN 'Asian'  
      WHEN race LIKE '%BLACK%' THEN 'Black'  
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'  
      WHEN race LIKE '%WHITE%' THEN 'White'  
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska  
Native'  
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native  
Hawaiian/Pacific Islander'  
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO  
ANSWER') THEN 'Unknown/Declined'  
      ELSE 'Other'  
    END AS race_group  
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`  
)  
  
patient_demo AS (  
  -- Join in gender  
  SELECT  
    r.hadm_id,  
    r.subject_id,  
    r.race_group,  
    p.gender  
  FROM race_bucketed r  
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p  
    ON r.subject_id = p.subject_id  
  WHERE r.dischtime > r.admittime  
)  
  
...
```

```
mental_health_dx AS (  
  -- Flag diagnosis rows that fall in mental/behavioral health ICD ranges  
  -- ICD-9: codes 290-319 (Mental Disorders chapter)  
  -- ICD-10: codes starting with 'F' (Mental, Behavioral and Neurodevelopmental disorders)  
  SELECT DISTINCT  
    hadm_id  
  FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`  
  WHERE  
    (icd_version = 9  
     AND SAFE_CAST(LEFT(icd_code, 3) AS INT64) BETWEEN 290 AND 319)  
    OR  
    (icd_version = 10  
     AND LEFT(icd_code, 1) = 'F')  
)  
  
admission_flagged AS (  
  -- Join demographics to the mental health flag  
  SELECT  
    d.hadm_id,  
    d.gender,  
    d.race_group,  
    CASE WHEN m.hadm_id IS NOT NULL THEN 1 ELSE 0 END AS has_mental_health_dx  
  FROM patient_demo d  
  LEFT JOIN mental_health_dx m ON d.hadm_id = m.hadm_id  
)  
  
-- Aggregate into final result of mental health diagnosis rate by gender and race  
SELECT  
  gender,  
  race_group,  
  COUNT(*) AS n_admissions,  
  SUM(has_mental_health_dx) AS n_with_mental_health_dx,  
  ROUND(AVG(has_mental_health_dx), 3) AS pct_with_mental_health_dx  
FROM admission_flagged  
GROUP BY gender, race_group  
ORDER BY pct_with_mental_health_dx DESC
```

Query 4: Visualization & Interpretation

I wanted to investigate if women were diagnosed with more mental illness than men. It seems that men are overwhelmingly more likely to be diagnosed with a mental illness than women, with exception of white women. Interestingly enough, minority men have the highest rate. This is certainly worth more research, coupled with the recent findings of minority men not being administered pain medication.



Query 4: Output

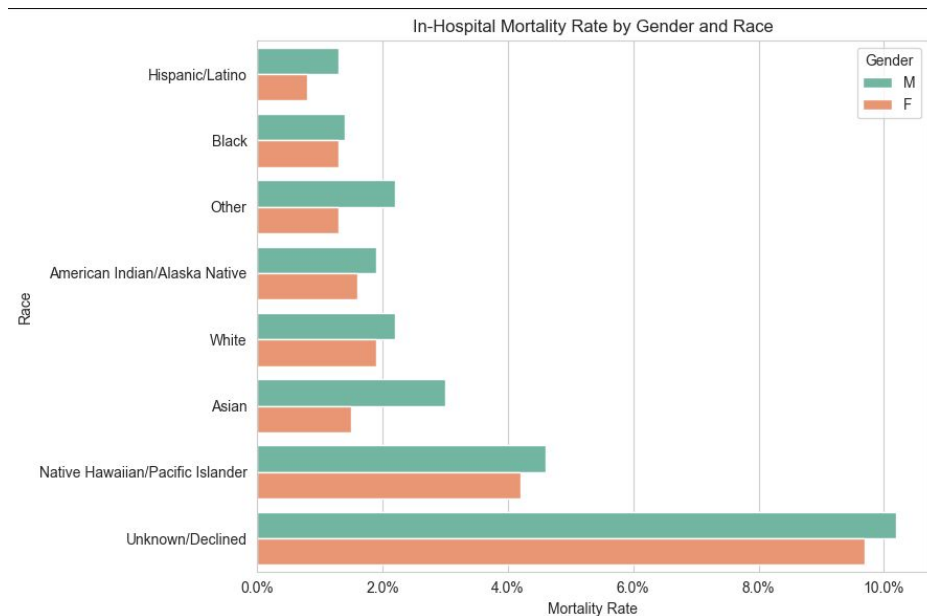
	gender	race_group	n_admissions	n_with_mental_health_dx	pct_with_mental_health_dx
0	M	Native Hawaiian/Pacific Islander	305	177	0.580
1	M	Black	35582	17598	0.495
2	M	American Indian/Alaska Native	567	279	0.492
3	M	Hispanic/Latino	14774	7168	0.485
4	F	Native Hawaiian/Pacific Islander	189	88	0.466
5	F	White	180105	83401	0.463
6	M	Unknown/Declined	10649	4779	0.449
7	M	White	180306	80454	0.446
8	F	Unknown/Declined	8831	3647	0.413
9	M	Other	11052	4465	0.404
10	F	American Indian/Alaska Native	680	261	0.384
11	F	Hispanic/Latino	17428	6668	0.383
12	F	Black	53455	20026	0.375
13	F	Other	12182	4454	0.366
14	M	Asian	8603	2514	0.292
15	F	Asian	11140	2511	0.225

Query 5: Mortality rate by gender and race

```
WITH race_bucketed AS (  
  -- Consolidate the raw race categories into more general groups  
  SELECT  
    hadm_id,  
    subject_id,  
    admittance,  
    dischtime,  
    hospital_expire_flag,  
    CASE  
      WHEN race LIKE '%ASIAN%' THEN 'Asian'  
      WHEN race LIKE '%BLACK%' THEN 'Black'  
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'  
      WHEN race LIKE '%WHITE%' THEN 'White'  
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska  
Native'  
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native  
Hawaiian/Pacific Islander'  
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO  
ANSWER') THEN 'Unknown/Declined'  
      ELSE 'Other'  
    END AS race_group  
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`  
)  
  
patient_demo AS (  
  -- Join in gender  
  SELECT  
    r.hadm_id,  
    r.subject_id,  
    r.race_group,  
    r.hospital_expire_flag,  
    p.gender  
  FROM race_bucketed r  
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p  
    ON r.subject_id = p.subject_id  
  WHERE r.dischtime > r.admittime  
)  
  
-- Aggregate into final result: in-hospital mortality rate by gender and race  
SELECT  
  gender,  
  race_group,  
  COUNT(*) AS n_admissions,  
  SUM(hospital_expire_flag) AS n_deaths,  
  ROUND(AVG(hospital_expire_flag), 3) AS mortality_rate  
FROM patient_demo  
GROUP BY gender, race_group  
ORDER BY mortality_rate DESC
```

Query 5: Visualization & Interpretation

Not looking at declined race, we can see once again that native hawaiian/pacific islanders are in the worst shape. They have a much higher rate of in-hospital mortality rate than other races. Asian men have a very high rate compared to their female counterparts, doubling asian women rates. Overall, men have a higher rate than women for all races.



Query 5: Output

	gender	race_group	n_admissions	n_deaths	mortality_rate
0	M	Unknown/Declined	10649	1091	0.102
1	F	Unknown/Declined	8831	860	0.097
2	M	Native Hawaiian/Pacific Islander	305	14	0.046
3	F	Native Hawaiian/Pacific Islander	189	8	0.042
4	M	Asian	8603	258	0.030
5	M	White	180306	4033	0.022
6	M	Other	11052	245	0.022
7	F	White	180105	3333	0.019
8	M	American Indian/Alaska Native	567	11	0.019
9	F	American Indian/Alaska Native	680	11	0.016
10	F	Asian	11140	171	0.015
11	M	Black	35582	495	0.014
12	F	Black	53455	671	0.013
13	M	Hispanic/Latino	14774	195	0.013
14	F	Other	12182	161	0.013
15	F	Hispanic/Latino	17428	143	0.008

Query 6: Rate of insurance types by unit

```
WITH icu_admissions AS (
  SELECT
    i.first_careunit,
    a.insurance
  FROM `physionet-data.mimiciv_3_1_icu.icustays` i
  JOIN `physionet-data.mimiciv_3_1_hosp.admissions` a
    ON i.hadm_id = a.hadm_id
),

unit_insurance_counts AS (
  SELECT
    first_careunit,
    insurance,
    COUNT(*) AS n_stays
  FROM icu_admissions
  GROUP BY first_careunit, insurance
),

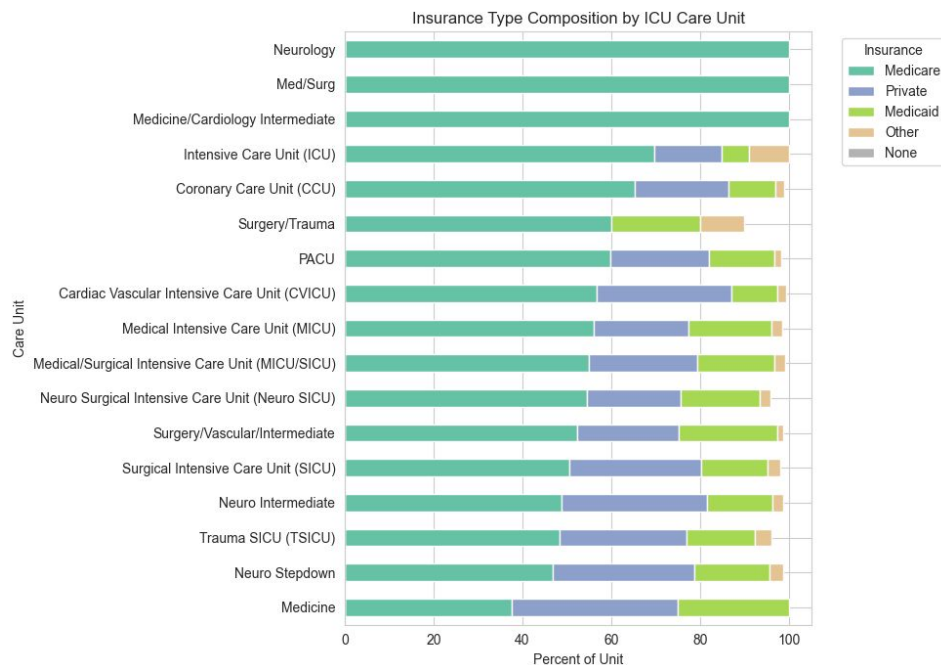
unit_insurance_pct AS (
  SELECT
    first_careunit,
    insurance,
    ROUND(
      n_stays / (SELECT SUM(n_stays) FROM unit_insurance_counts u2 WHERE
u2.first_careunit = u.first_careunit) * 100,
      1
    ) AS pct_of_unit
  FROM unit_insurance_counts u
),

pivoted AS (
  SELECT *
  FROM unit_insurance_pct
  PIVOT (
    SUM(pct_of_unit)
    FOR insurance IN ('Medicare', 'Private', 'Medicaid', 'Other', 'None')
  )
)

SELECT
  first_careunit,
  IFNULL(Medicare, 0.0) AS Medicare,
  IFNULL(Private, 0.0) AS Private,
  IFNULL(Medicaid, 0.0) AS Medicaid,
  IFNULL(Other, 0.0) AS Other,
  IFNULL(None, 0.0) AS None
FROM pivoted
ORDER BY first_careunit
```

Query 6: Visualization & Interpretation

Medicare has the highest rate for virtually every ICU unit (with exception of Medicine). It is worth investigating more why Medicare makes up 100% of insurance for Neuro, MedSurg, and Intermediate unites. Private insurance comes in second, as expected.



Query 6: Output

	first_careunit	Medicare	Private	Medicaid	Other	None
0	Cardiac Vascular Intensive Care Unit (CVICU)	56.8	30.2	10.4	1.9	0.0
1	Coronary Care Unit (CCU)	65.3	21.1	10.6	1.9	0.0
2	Intensive Care Unit (ICU)	69.7	15.2	6.1	9.1	0.0
3	Med/Surg	100.0	0.0	0.0	0.0	0.0
4	Medical Intensive Care Unit (MICU)	56.1	21.3	18.7	2.3	0.0
5	Medical/Surgical Intensive Care Unit (MICU/SICU)	55.0	24.3	17.5	2.3	0.0
6	Medicine	37.5	37.5	25.0	0.0	0.0
7	Medicine/Cardiology Intermediate	100.0	0.0	0.0	0.0	0.0
8	Neuro Intermediate	48.9	32.6	14.8	2.4	0.0
9	Neuro Stepdown	46.8	31.9	17.0	3.0	0.0
10	Neuro Surgical Intensive Care Unit (Neuro SICU)	54.5	21.1	17.9	2.4	0.0
11	Neurology	100.0	0.0	0.0	0.0	0.0
12	PACU	59.8	22.1	14.8	1.6	0.0
13	Surgery/Trauma	60.0	0.0	20.0	10.0	0.0
14	Surgery/Vascular/Intermediate	52.4	22.8	22.1	1.4	0.0
15	Surgical Intensive Care Unit (SICU)	50.6	29.7	14.8	3.0	0.0
16	Trauma SICU (TSICU)	48.3	28.7	15.4	3.7	0.0

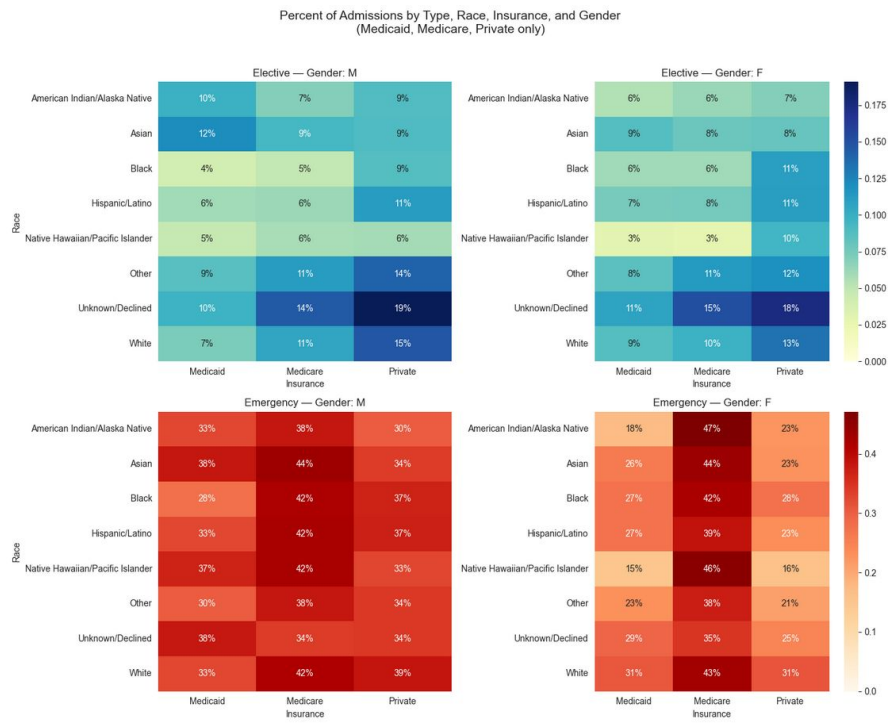
Query 7: Admission type by gender, race, and insurance type

```
WITH race_bucketed AS (  
  SELECT  
    hadm_id,  
    subject_id,  
    insurance,  
    admittance,  
    dischtime,  
    CASE  
      WHEN admission_type LIKE '%EMER%' THEN 'Emergency'  
      WHEN admission_type = 'URGENT' THEN 'Urgent'  
      WHEN admission_type = 'ELECTIVE' THEN 'Elective'  
      WHEN admission_type = 'SURGICAL SAME DAY ADMISSION' THEN 'Elective'  
      WHEN admission_type LIKE '%OBSERVATION%' THEN 'Observation'  
      ELSE 'Other'  
    END AS admission_type_group,  
    CASE  
      WHEN race LIKE '%ASIAN%' THEN 'Asian'  
      WHEN race LIKE '%BLACK%' THEN 'Black'  
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'  
      WHEN race LIKE '%WHITE%' THEN 'White'  
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska  
Native'  
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native  
Hawaiian/Pacific Islander'  
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO  
ANSWER') THEN 'Unknown/Declined'  
      ELSE 'Other'  
    END AS race_group  
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`  
)  
  
patient_demo AS (  
  SELECT  
    r.hadm_id,  
    r.insurance,  
    r.race_group,  
    r.admission_type_group,  
    p.gender  
  FROM race_bucketed r  
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p  
    ON r.subject_id = p.subject_id  
  WHERE r.dischtime > r.admittime  
)
```

```
group_totals AS (  
  SELECT  
    gender,  
    insurance,  
    race_group,  
    COUNT(*) AS total_admissions  
  FROM patient_demo  
  GROUP BY gender, insurance, race_group  
)  
  
type_counts AS (  
  SELECT  
    gender,  
    insurance,  
    race_group,  
    admission_type_group,  
    COUNT(*) AS n_admissions  
  FROM patient_demo  
  GROUP BY gender, insurance, race_group, admission_type_group  
)  
  
SELECT  
  t.gender,  
  t.insurance,  
  t.race_group,  
  t.admission_type_group,  
  t.n_admissions,  
  g.total_admissions,  
  ROUND(t.n_admissions / g.total_admissions * 100, 1) AS  
  pct_of_group  
FROM type_counts t  
JOIN group_totals g  
  ON t.gender = g.gender  
  AND t.insurance = g.insurance  
  AND t.race_group = g.race_group  
ORDER BY t.admission_type_group DESC
```

Query 7: Visualization & Interpretation

Private insurance makes up the largest chunk of elective admissions, while medicare makes up the largest chunk of emergency admissions. This is consistent with socioeconomically disadvantaged people not having access to elective or preventative care, and vice versa.



Query 7: Output

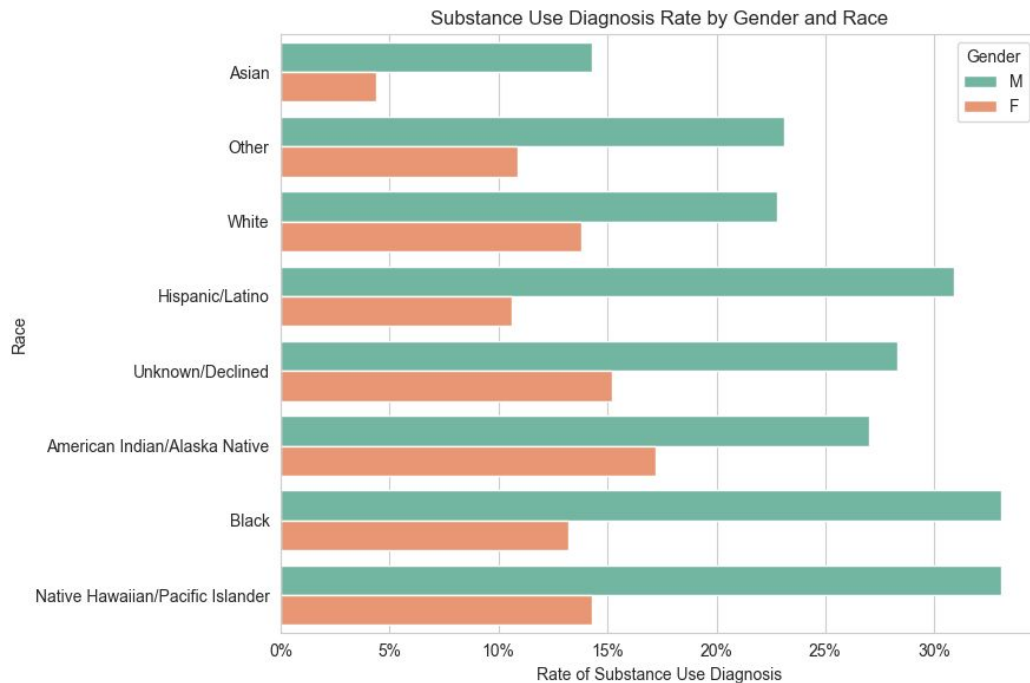
	gender	insurance	race_group	admission_type_group	n_admissions	total_admissions	pct_of_group
0	F	Medicaid	Asian	Urgent	868	3308	26.2
1	F	No charge	Unknown/Declined	Urgent	1	22	4.5
2	F	No charge	Other	Urgent	1	4	25.0
3	M	No charge	Black	Urgent	3	37	8.1
4	F	Medicare	White	Urgent	6345	90618	7.0
...	...	...	...	...	...	...	...
288	F	Other	American Indian/Alaska Native	Elective	2	28	7.1
289	M	Other	American Indian/Alaska Native	Elective	1	38	2.6
290	F	Private	Unknown/Declined	Elective	525	2996	17.5
291	F	Medicare	White	Elective	8864	90618	9.8
292	F	No charge	Unknown/Declined	Elective	18	22	81.8
293	F	No charge	Unknown/Declined	Elective	18	22	81.8

Query 8: Drug use by race and gender

```
WITH race_bucketed AS (  
  -- Consolidate the raw race categories into more general groups  
  SELECT  
    hadm_id,  
    subject_id,  
    admittance,  
    discharge,  
    CASE  
      WHEN race LIKE '%ASIAN%' THEN 'Asian'  
      WHEN race LIKE '%BLACK%' THEN 'Black'  
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'  
      WHEN race LIKE '%WHITE%' THEN 'White'  
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska  
Native'  
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native  
Hawaiian/Pacific Islander'  
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO  
ANSWER') THEN 'Unknown/Declined'  
      ELSE 'Other'  
    END AS race_group  
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`  
)  
  
patient_demo AS (  
  -- Join in gender  
  SELECT  
    r.hadm_id,  
    r.race_group,  
    p.gender  
  FROM race_bucketed r  
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p  
    ON r.subject_id = p.subject_id  
  WHERE r.discharge > r.admittance  
)  
  
substance_use_dx AS (  
  -- Flag diagnosis rows for substance-use-related disorders  
  -- ICD-9: 291-292 (alcohol/drug-induced psychoses), 303-305 (dependence/abuse)  
  -- ICD-10: F10-F19 (mental/behavioral disorders due to psychoactive substance use)  
  SELECT DISTINCT  
    hadm_id  
  FROM `physionet-data.mimiciv_3_1_hosp.diagnoses_icd`  
  WHERE  
    (icd_version = 9  
      AND (  
        SAFE_CAST(LEFT(icd_code, 3) AS INT64) BETWEEN 291 AND 292  
        OR SAFE_CAST(LEFT(icd_code, 3) AS INT64) BETWEEN 303 AND 305  
      ))  
    OR  
    (icd_version = 10  
      AND LEFT(icd_code, 3) BETWEEN 'F10' AND 'F19')  
)  
)  
  
admission_flagged AS (  
  -- Join demographics to substance use flag (LEFT JOIN keeps admissions with no such diagnosis)  
  SELECT  
    d.hadm_id,  
    d.gender,  
    d.race_group,  
    CASE WHEN s.hadm_id IS NOT NULL THEN 1 ELSE 0 END AS has_substance_use_dx  
  FROM patient_demo d  
  LEFT JOIN substance_use_dx s ON d.hadm_id = s.hadm_id  
)  
  
-- Aggregate into final result: substance use diagnosis rate by gender and race  
SELECT  
  gender,  
  race_group,  
  COUNT(*) AS n_admissions,  
  SUM(has_substance_use_dx) AS n_with_substance_use_dx,  
  ROUND(AVG(has_substance_use_dx), 3) AS pct_with_substance_use_dx  
FROM admission_flagged  
GROUP BY gender, race_group  
ORDER BY pct_with_substance_use_dx DESC
```

Query 8: Visualization & Interpretation

Men had overwhelmingly more recorded drug use than women, with the highest rates amongst minorities. Asians had the lowest rates of drug use overall.



Query 8: Output

	gender	race_group	n_admissions	n_with_substance_use_dx	pct_with_substance_use_dx
0	M	Black	35582	11780	0.331
1	M	Native Hawaiian/Pacific Islander	305	101	0.331
2	M	Hispanic/Latino	14774	4572	0.309
3	M	Unknown/Declined	10649	3014	0.283
4	M	American Indian/Alaska Native	567	153	0.270
5	M	Other	11052	2553	0.231
6	M	White	180306	41118	0.228
7	F	American Indian/Alaska Native	680	117	0.172
8	F	Unknown/Declined	8831	1339	0.152
9	M	Asian	8603	1231	0.143
10	F	Native Hawaiian/Pacific Islander	189	27	0.143
11	F	White	180105	24917	0.138
12	F	Black	53455	7050	0.132
13	F	Other	12182	1329	0.109
14	F	Hispanic/Latino	17428	1855	0.106
15	F	Asian	11140	488	0.044

Query 9: Discharge disposition by race, gender, and insurance

```

WITH race_bucketed AS (
  -- Consolidate the raw race categories into more general groups
  SELECT
    hadm_id,
    subject_id,
    insurance,
    admittance,
    disctime,
    discharge_location,
    CASE
      WHEN race LIKE '%ASIAN%' THEN 'Asian'
      WHEN race LIKE '%BLACK%' THEN 'Black'
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'
      WHEN race LIKE '%WHITE%' THEN 'White'
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska
Native'
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native
Hawaiian/Pacific Islander'
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO
ANSWER') THEN 'Unknown/Declined'
      ELSE 'Other'
    END AS race_group
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`
),

patient_demo AS (
  -- Join in gender, bucket discharge location
  SELECT
    r.hadm_id,
    r.insurance,
    r.race_group,
    p.gender,
    CASE
      WHEN r.discharge_location = 'HOME' THEN 'Home (self)'
      WHEN r.discharge_location = 'HOME HEALTH CARE' THEN 'Home (with care)'
      WHEN r.discharge_location IN ('SKILLED NURSING FACILITY', 'REHAB',
'CHRONIC/LONG TERM ACUTE CARE', 'ASSISTED LIVING') THEN 'Facility care'
      WHEN r.discharge_location = 'HOSPICE' THEN 'Hospice'
      WHEN r.discharge_location = 'DIED' THEN 'Died'
      WHEN r.discharge_location = 'AGAINST ADVICE' THEN 'Against medical advice'
      WHEN r.discharge_location IN ('ACUTE HOSPITAL', 'PSYCH FACILITY', 'OTHER
FACILITY', 'HEALTHCARE FACILITY') THEN 'Transferred'
      WHEN r.discharge_location IS NULL THEN 'Unknown/Not recorded'
      ELSE 'Other/Unknown'
    END AS disposition_group
  FROM race_bucketed r
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p
    ON r.subject_id = p.subject_id
  WHERE r.disctime > r.admittime
),

group_totals AS (
  SELECT
    gender,
    insurance,
    race_group,
    COUNT(*) AS total_admissions
  FROM patient_demo
  GROUP BY gender, insurance, race_group
),

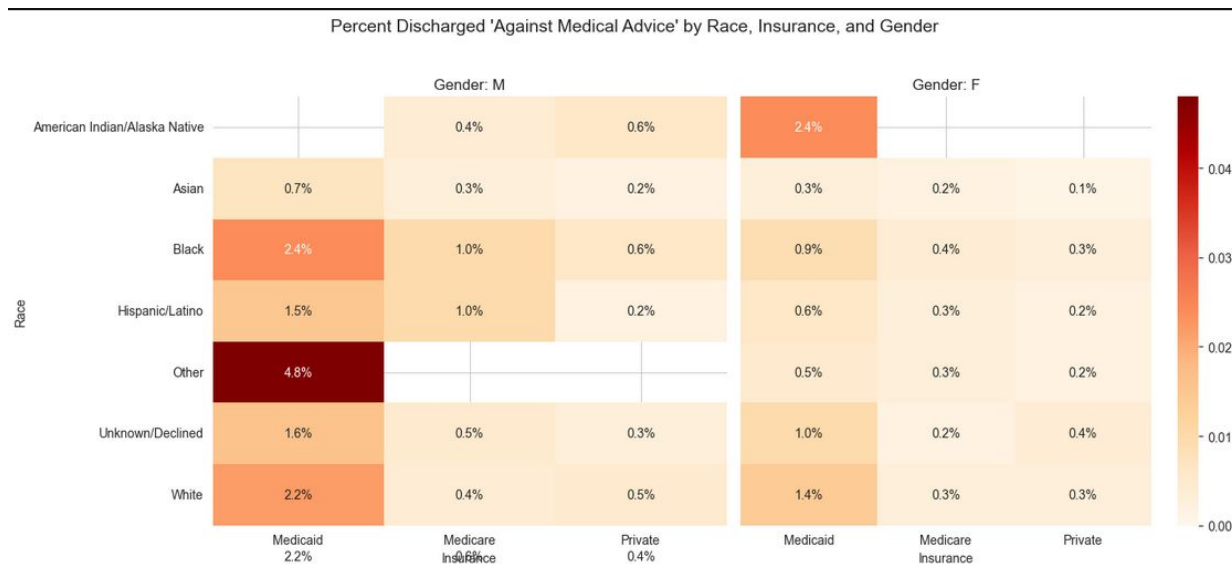
disposition_counts AS (
  SELECT
    gender,
    insurance,
    race_group,
    disposition_group,
    COUNT(*) AS n_admissions
  FROM patient_demo
  GROUP BY gender, insurance, race_group, disposition_group
)

SELECT
  d.gender,
  d.insurance,
  d.race_group,
  d.disposition_group,
  d.n_admissions,
  g.total_admissions,
  ROUND(d.n_admissions / g.total_admissions * 100, 1) AS pct_of_group
FROM disposition_counts d
JOIN group_totals g
  ON d.gender = g.gender
  AND d.insurance = g.insurance
  AND d.race_group = g.race_group
ORDER BY d.disposition_group DESC

```

Query 9: Visualization & Interpretation

Patients on Medicaid left against medical advice more than any other insurance type. Black men had the highest instance of this, with men in general having a much higher rate of leaving AMA than women in all cases.



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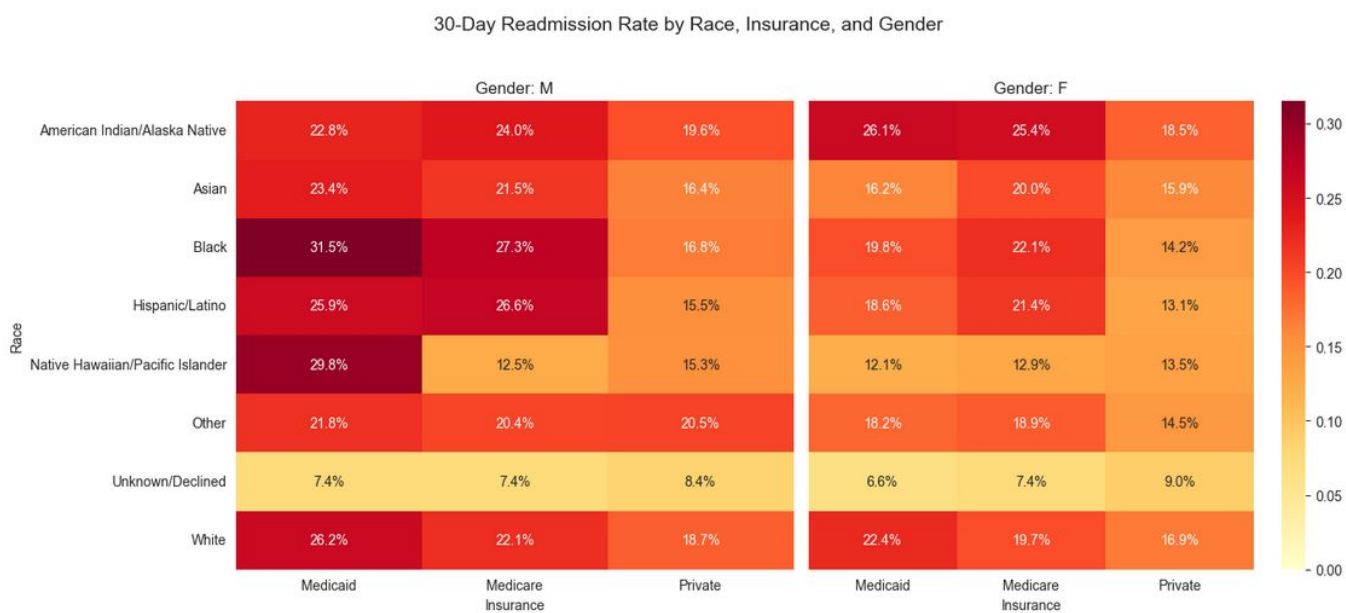
Query 10: Discharge disposition by race, gender, and insurance

```
WITH race_bucketed AS (  
  SELECT  
    hadm_id,  
    subject_id,  
    insurance,  
    admittance,  
    discharge,  
    CASE  
      WHEN race LIKE '%ASIAN%' THEN 'Asian'  
      WHEN race LIKE '%BLACK%' THEN 'Black'  
      WHEN race LIKE '%HISPANIC%' THEN 'Hispanic/Latino'  
      WHEN race LIKE '%WHITE%' THEN 'White'  
      WHEN race = 'AMERICAN INDIAN/ALASKA NATIVE' THEN 'American Indian/Alaska  
Native'  
      WHEN race = 'NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER' THEN 'Native  
Hawaiian/Pacific Islander'  
      WHEN race IN ('UNKNOWN', 'UNABLE TO OBTAIN', 'PATIENT DECLINED TO  
ANSWER') THEN 'Unknown/Declined'  
      ELSE 'Other'  
    END AS race_group  
  FROM `physionet-data.mimiciv_3_1_hosp.admissions`  
)
```

```
patient_demo AS (  
  SELECT  
    r.hadm_id,  
    r.subject_id,  
    r.insurance,  
    r.race_group,  
    r.admittime,  
    r.dischtime,  
    p.gender  
  FROM race_bucketed r  
  JOIN `physionet-data.mimiciv_3_1_hosp.patients` p  
    ON r.subject_id = p.subject_id  
  WHERE r.dischtime > r.admittime  
)  
  
readmission_flagged AS (  
  -- Self-join: match each admission to the patient's next-later admission  
  SELECT  
    d.hadm_id,  
    d.gender,  
    d.insurance,  
    d.race_group,  
    CASE  
      WHEN MIN(d2.admittime) IS NOT NULL  
        AND DATETIME_DIFF(MIN(d2.admittime), d.dischtime, DAY) <= 30  
      THEN 1  
      ELSE 0  
    END AS readmitted_30day  
  FROM patient_demo d  
  LEFT JOIN patient_demo d2  
    ON d.subject_id = d2.subject_id  
    AND d2.admittime > d.dischtime  
  GROUP BY d.hadm_id, d.gender, d.insurance, d.race_group, d.dischtime  
)  
  
SELECT  
  gender,  
  insurance,  
  race_group,  
  COUNT(*) AS n_admissions,  
  SUM(readmitted_30day) AS n_readmissions,  
  ROUND(AVG(readmitted_30day), 3) AS readmission_rate_30day  
FROM readmission_flagged  
GROUP BY gender, insurance, race_group  
ORDER BY readmission_rate_30day DESC
```

Query 10: Visualization & Interpretation

We see similar results to query 9, where the 30 day readmission rate is higher for men in most cases, and especially high for men on Medicaid, and extremely high for minorities in this category. It is worth more research on if this is correlated to leaving AMA as previously seen.



Query 9: Output

[67]:

	gender	insurance	race_group	n_admissions	n_readmissions	readmission_rate_30day
0	F	None	American Indian/Alaska Native	4	2	0.500
1	F	Other	Native Hawaiian/Pacific Islander	8	4	0.500
2	F	No charge	Hispanic/Latino	9	3	0.333
3	F	Other	American Indian/Alaska Native	28	9	0.321
4	M	Medicaid	Black	10937	3443	0.315
...	...	...	...	...	...	...
87	F	None	Unknown/Declined	338	22	0.065
88	M	None	Unknown/Declined	689	34	0.049
89	M	Other	Native Hawaiian/Pacific Islander	6	0	0.000
90	F	None	Native Hawaiian/Pacific Islander	4	0	0.000
91	F	No charge	Other	4	0	0.000

92 rows × 6 columns

Story - Bonus Credit

Each query has investigated MIMIC IV data with special regard to socioeconomic factors race, gender, and insurance type. This forms the basis of a complex and comprehensive research project evaluating why we see certain results that relate especially to men, minorities, and patients on Medicaid.

The story shows that patients on Medicaid have worse experiences and outcomes in the hospital system. Further it shows that minorities, especially minority men also have worse outcomes compared to their female and white counterparts.